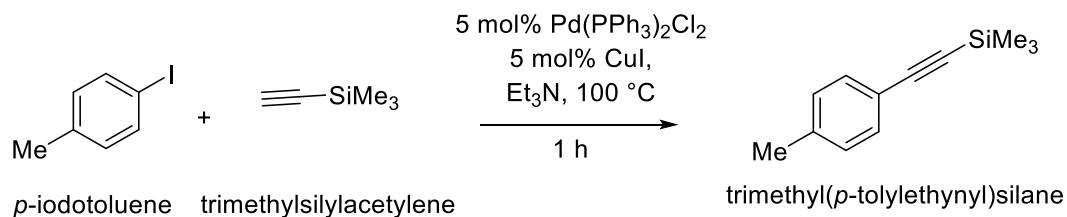


## CH4203: Organic Synthesis Laboratory

**Expt. No. 4:** Sonogashira Cross-Coupling of Iodoarene with Terminal Alkyne: Synthesis of Disubstituted Alkynes

**Aim and Objectives:** In this lab, we will learn **Sonogashira Coupling** which is nothing but a **Cross-Coupling** reaction and we will make a disubstituted alkyne from 4-iodotoluene and trimethylsilylacetylene.



### **Apparatus Requirement:**

Sealed tube; Needle-Syringe; Round Bottom (RB) flask; Conical flask; Measuring cylinder; Separating funnel; Funnel.

### **Chemicals Requirement:**

1. 4-Iodotoluene
2. Trimethylsilylacetylene
3. Triethyl amine
4. Pd(PPh<sub>3</sub>)<sub>2</sub>Cl<sub>2</sub>
5. Copper iodide

### **Molecular Wt. and Physical Details of Reagents:**

Molecular formula of 4-Iodotoluene = C<sub>7</sub>H<sub>7</sub>I

Molecular weight of 4-Iodotoluene = 218.0375 g/mole

Molecular formula of Trimethylsilylacetylene = C<sub>5</sub>H<sub>10</sub>Si

Molecular weight of Trimethylsilylacetylene = 98.22 g/mole

Density of Trimethylsilylacetylene = 0.695 g/ml

Molecular weight of Pd(PPh<sub>3</sub>)<sub>2</sub>Cl<sub>2</sub> = 701.89 g/mole

Molecular weight of CuI = 190.45 g/mole

## Experimental Procedure:

In an over dried sealed tube fitted with a magnetic stirrer, 1 gm (4.58 mmol) of 4-Iodotoluene was taken in 25 ml of triethylamine and it was stirred for 2 minutes.

To this solution was added 1.62 mL (11.45 mmol) of trimethylsilyl acetylene at room temperature using a syringe and it was stirred for 2 minutes.

Next, the mixture was degassed using N<sub>2</sub> balloon, needle and septum for 10 minutes in stirring condition to make sure that the reaction is under oxygen-free atmosphere.

Following de-gas of the reaction mixture, 42 mg (5 mol% with respect to 4-iodotoluene) of CuI (as co-catalyst to activate the terminal alkyne) was added quickly to the mixture under N<sub>2</sub> atmosphere and the tube was sealed using the cap.

After that, 160 mg (5 mol% with respect to 4-iodotoluene) Pd(PPh<sub>3</sub>)<sub>2</sub>Cl<sub>2</sub> (catalyst for the cross-coupling reaction) was added quickly to the reaction mixture without much exposing to air. This Pd(II) can reduced in the reaction mixture.

Once all the reagents were added, the sealed tube was dipped in a pre-heated oil bath maintaining temperature at 100 °C. After 1 h, TLC was checked to confirm the completion of the reaction.

Upon completion of the reaction (as monitored by running TLC), the reaction mixture was transferred to a Round Bottom flask and most of the triethylamine was evaporated under the reduced pressure.

Next, the reaction mixture was diluted with 20 mL hexane and water (20 mL) and the Round Bottom flask was stirred vigorously. Then the layers were separated using separating funnel and the organic layer was collected in a conical flask. This was repeated two more time (20 mL X 2).

Finally, the organic layer was evaporated and column was done using 100-200 mess silica in n-hexane.

$R_f$  (in *n*-hexane) = 0.8

**Note:** Triethylamine may cause a skin allergy, itching and a skin rash. So, proper precaution have to be taken during the experiment.

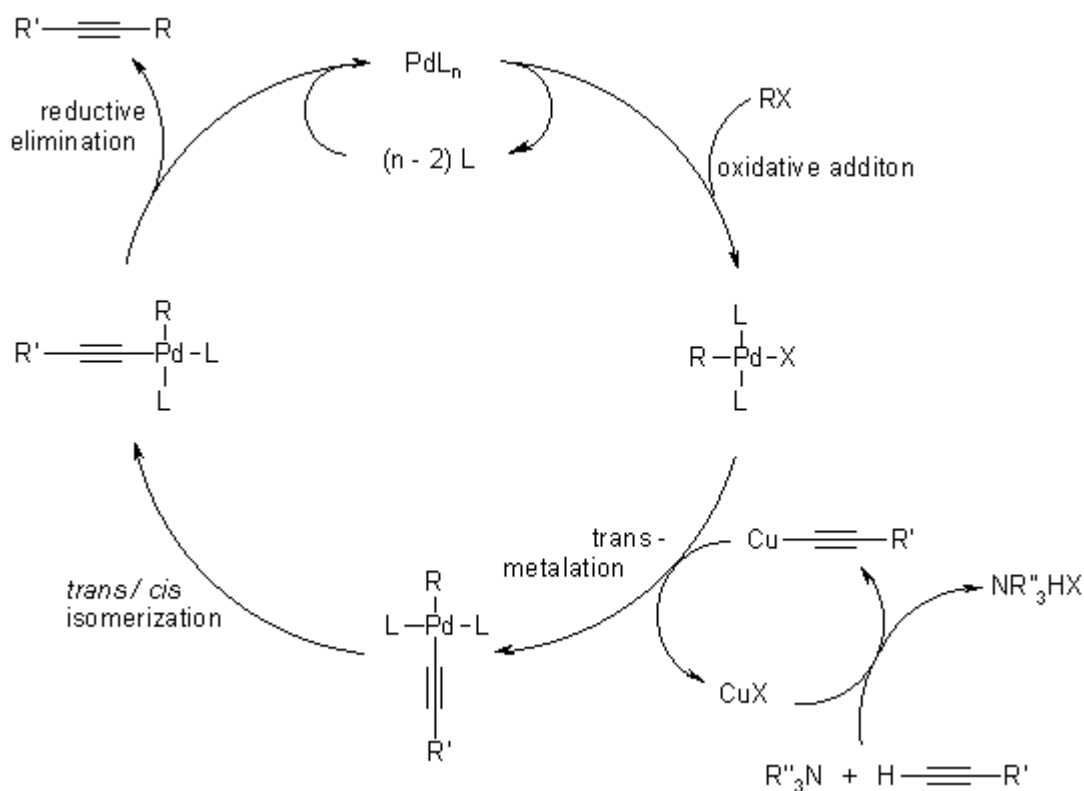
## Characterization of Disubstituted alkyne:

In summary, we have conducted **Sonogashira** Cross-Coupling for the Synthesis of Disubstituted alkyne.

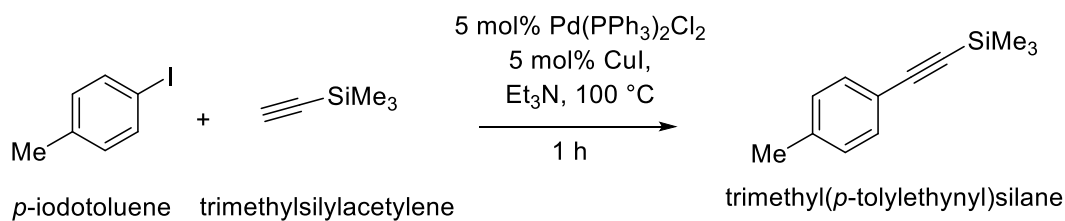
The final product was characterized by **IR** spectra, **<sup>1</sup>H NMR** spectra, **<sup>13</sup>C NMR** spectra, and **HRMS** spectroscopy.

## Reaction Mechanism:

Sonogashira Coupling for the synthesis of Disubstituted alkyne reaction mechanism (as model reaction)



## Calculations (mmol):



1.0 eq.

2.5 eq.

1.0 eq

MW: 218.04

MW: 98.22  
density = 0.695 gm/ml

MW: 188.345

**1000 mg**  
(4.58 mmol)

**1.62 mL**  
(11.46 mmol)

Theoretical Yield (100%)  
**862 mg**

### Results & Observation:

Yield: 819.5 mg

% of Yield: 95%

TLC System:  $R_f = 0.8$  (in *n*-hexane)

### Compound Characterization:

IR Frequencies:  $\text{cm}^{-1}$ <sup>1</sup>H-NMR: (400 MHz, CDCl<sub>3</sub>) δ $^{13}\text{C}$ -NMR: (126 MHz,  $\text{CDCl}_3$ )  $\delta$ 

### Conclusion:

In summary, we have done the Sonogashira Cross-Coupling for the synthesis of disubstituted alkyne. In particular, we have chosen 4-iodotoluene as haloarene and trimethylsilylacetylene as the coupling partner in the presence of Pd(0) as catalyst and CuI as co-catalyst.